

Numerical Analysis 2

Benabderrahmane Benyattou

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Chapter 1

Chapter 3 : Numerical methods for Ordinary Differential Equations, ODEs

1.1 Introduction to initial value problems

Definition 1 (*Ordinary Differential Equations*): Let $y(x)$ be a function defined on an interval $I \subseteq \mathbb{R}$ with values in the real or complex numbers or in the space $\mathbb{R}^m$, or $\mathbb{C}^m$. An ordinary differential equation (ODE) is an equation for $y(x)$ of the form

$$F\left(x, y(x), y'(x), \dots, y^{(n)}(x)\right)$$

Here F denotes an arbitrary function of its arguments.

The order n of a differential equation is the highest derivative which occurs. If $m > 1$, we talk about systems of differential equations.

Remark 2 A differential equation (DE), which is not ordinary, is called partial. These are equations or systems of equations, which involve partial derivatives with respect to several independent variables. While the functions in an ordinary differential equation may be dependent on additional parameters, derivatives are only taken with respect to one variable. Often, but not exclusively, this variable is time. This manuscript only deals with ordinary differential equations, and so the adjective will be omitted in the following.

Definition 3 An explicit differential equation of first order is a equation of the form

$$y'(x) = f(x, y(x)).$$



A differential equation of order n is called *explicit*, if it is of the form

$$y^{(n)}(x) = f\left(x, y(x), y'(x), \dots, y^{(n-1)}(x)\right).$$

Lemma 4 *Every initial differential equation (of arbitrary order) can be written as a system of first-order differential equations. If the equation is explicit, then the system is explicit.*

Proof. Let

$$y^{(n)}(x) = f\left(x, y(x), y'(x), \dots, y^{(n-1)}(x)\right).$$

We introduce the additional variables

$$z_1(x) = y(x), z_2(x) = y'(x), \dots, z_n(x) = y^{(n-1)}(x),$$

Then, the differential equation can be reformulated as the system

$$\begin{cases} z_1'(x) = z_2(x); \\ z_2'(x) = z_3(x); \\ \vdots \\ z_n'(x) = f(x, z_1, \dots, z_n). \end{cases}$$

Thus, we have the single vector equation

$$z'(x) = F(x, z(x)),$$

where

$$\begin{aligned} z(x) &= (z_1(x), \dots, z_n(x))^T, \\ F(x, z(x)) &= \begin{pmatrix} z_2(x) \\ z_3(x) \\ \vdots \\ f(x, z_1, \dots, z_n) \end{pmatrix}. \end{aligned}$$

■

Definition 5 (*Initial Value Problem*) : Given a point $(x_0, y_0) \in \mathbb{R} \times \mathbb{R}^m$. Then, an initial value problem (IVP) is defined as follows:

$$(IVP) : \begin{cases} \text{Find a function } y(x) \text{ such that} \\ y'(x) = f(x, y(x)) \\ y(x_0) = y_0 \end{cases}$$

Definition 6 (*Local Solution*) : A continuously differentiable function $y(x)$ with $y(x_0) = y_0$ is called a *local solution* of the (IVP), if there exists a neighborhood $I \subset \mathbb{R}$ of x_0 , such that $y(x)$ and $f(x, y(x))$ are defined and equation $y'(x) = f(x, y(x))$ holds for all $x \in I$.



Proposition 7 (Volterra integral equation) : Let f be continuous in both arguments. Then, the function $y(x)$ is a solution of the initial value problem (IVP) if and only if it is a solution of the Volterra integral equation (VIE)

$$y(x) = y_0 + \int_{x_0}^x f(t, y(t)) dt.$$

Example 8 Consider the simple problem

$$y' = -2xy, y(0) = 1.$$

Here $f(x, y) = -2xy, x_0 = 0, y_0 = 1$.

Example 9 Consider now the simple system of equations

$$\begin{aligned} y_1' &= -4y_1 + y_2, y_1(0) = 1; \\ y_2' &= y_1 - 4y_2, y_2(0) = 0. \end{aligned}$$

In this example, $y(x) = (y_1(x), y_2(x))^T$ is a vector of two components, and $f(x, y)$ is defined by

$$f(x, y) = (f_1(x, y(x)), f_2(x, y(x)))^T = (-4y_1 + y_2, y_1 - 4y_2)^T,$$

the initial point is still $x_0 = 0$, but the initial value is now the vector $y_0 = (1, 0)^T$.

Example 10 What about the second-order equation

$$y'' + 3y' + y = 0, y(0) = 1, y'(0) = 0?$$

It does not appear to fit the standard form for our initial value problems, yet we can put it into this form without a lot of work. Define the vector

$$z(x) = (z_1(x), z_2(x))^T, \text{ where } z_1(x) = y(x), z_2(x) = y'(x).$$

Thus we have the first-order system,

$$\begin{aligned} z_1'(x) &= z_2(x), z_1(0) = 1; \\ z_2'(x) &= -3z_2(x) - z_1(x), z_2(0) = 0. \end{aligned}$$

which can be written as follows

$$\begin{cases} z'(x) = f(x, z(x)) \\ z(0) = z_0, \end{cases}$$

where

$$z'(x) = \begin{pmatrix} z_1'(x) \\ z_2'(x) \end{pmatrix}, f(x, z(x)) = \begin{pmatrix} z_2(x) \\ -3z_2(x) - z_1(x) \end{pmatrix}, z_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$



Continuity of f guarantees local existence of solutions but not uniqueness. We need an extra assumption on f to ensure uniqueness of solutions.

Definition 11 A function f defined on $[x_0, b] \times \mathbb{R}^m$ is uniformly Lipschitz in y (with respect to some norm $\|\cdot\|$), if there is a constant $L \geq 0$ such that

$$\|f(x, y_1) - f(x, y_2)\| \leq L \|y_1 - y_2\|,$$

for all $x \in [x_0, b]$ and all $y_1, y_2 \in \mathbb{R}^m$.

Note that if $f \in C^1(\mathbb{R} \times \mathbb{R}^m)$, the Lipschitz condition is equivalent to

$$\exists L \geq 0 : \max_{x \in [x_0, b]} \|D_y f(x, y)\| \leq L, \text{ for all } y \in \mathbb{R}^m.$$

Example 12 Consider the two initial value problems

$$y' = 4y - e^{-x}, y(0) = 1,$$

and

$$y' = -(1 + x^2)y + \sin x, y(0) = 1.$$

We have

$$|f(x, y_1) - f(x, y_2)| = |4y_1 - 4y_2| \leq 4|y_1 - y_2|, \text{ and}$$

Hence, in this case f is Lipschitz in y with constant $L = 4$.

On the other hand, we have

$$\begin{aligned} |f(x, y_1) - f(x, y_2)| &= |-(1 + x^2)y_1 + \sin x + (1 + x^2)y_2 - \sin x| \\ &\leq (1 + x^2)|y_1 - y_2| \leq 2|y_1 - y_2|, \text{ for all } x \in [0, 1]. \end{aligned}$$

Therefore, this f is Lipschitz in y with constant $L = 2$.

Example 13 Consider the initial value problem

$$\begin{cases} y''' + 2y + \cos y' + e^x = 0, x \in [0, 2] \\ y(0) = y'(0) = 0, y''(0) = 1. \end{cases}$$

Let's define the vector

$$z(x) = (z_1(x), z_2(x), z_3(x))^T, \text{ with } z_1(x) = y(x), z_2(x) = y'(x), z_3(x) = y''(x).$$

Thus we have the first-order system,

$$\begin{aligned} z_1'(x) &= z_2(x), \quad z_1(0) = 0; \\ z_2'(x) &= z_3(x), \quad z_2(0) = 0; \\ z_3'(x) &= e^x - 2z_1(x) - \cos z_2(x), \quad z_3(0) = 1 \end{aligned}$$



which can be written as follows

$$\begin{cases} z'(x) = f(x, z(x)), f : \mathbb{R}^4 \longrightarrow \mathbb{R}^3 \\ z(0) = z_0, \end{cases}$$

where

$$z'(x) = \begin{pmatrix} z'_1(x) \\ z'_2(x) \\ z'_3(x) \end{pmatrix}, f(x, z(x)) = \begin{pmatrix} f_1(x, z(x)) \\ f_2(x, z(x)) \\ f_3(x, z(x)) \end{pmatrix} = \begin{pmatrix} z_2(x) \\ z_3(x) \\ -2z_1(x) - \cos z_2(x) - e^x \end{pmatrix}, z_0 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

So

$$D_z f(x, z) = \nabla_z f = \begin{pmatrix} \frac{\partial f_1}{\partial z_1} & \frac{\partial f_1}{\partial z_2} & \frac{\partial f_1}{\partial z_3} \\ \frac{\partial f_2}{\partial z_1} & \frac{\partial f_2}{\partial z_2} & \frac{\partial f_2}{\partial z_3} \\ \frac{\partial f_3}{\partial z_1} & \frac{\partial f_3}{\partial z_2} & \frac{\partial f_3}{\partial z_3} \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -2 & \sin z_2(x) & 0 \end{pmatrix}.$$

So,

$$\max_{x \in [0, 2]} \|D_z f(x, z)\|_p \leq L, p = 1, 2, \infty,$$

which proves that f is Lipschitz in z .

We now state a fundamental theorem of existence and uniqueness of solutions of the (IVP).

Theorem 14 (Existence and Uniqueness). *Let*

$$D = \{(x, y) : x \in [x_0, b], y \in \mathbb{R}^m\}.$$

If f is continuous in D and uniformly Lipschitz in y , the (IVP) has a unique solution for each $y_0 \in \mathbb{R}^n$.

Example 15 Consider

$$y' = 1 + \sin xy, 0 \leq x \leq 1, y \in \mathbb{R}.$$

To compute the Lipschitz constant L , we have

$$\left| \frac{\partial f}{\partial y} \right| = |x \cos xy| \leq 1 = L.$$

Since f is continuous in $[0, 1] \times \mathbb{R}$, thus, for any (x_0, y_0) there is a unique solution $y(x)$ to the associated initial value problem.

The numerical methods we will study next deal with the more general and important case when f depends on the unknown y . The methods produce an approximation to the exact solution of (IVP) (assuming uniqueness) at a set of discrete points

$$0 = x_0 < x_1 < \dots < x_n \leq b.$$

For simplicity in the presentation, we will assume these points are equispaced,

$$x_k = x_0 + k \frac{b - x_0}{n}, k = 0, 1, \dots, n,$$

but they do not have to be. We will write a numerical method for an (IVP) ε an algorithm to go from one discrete time, x_k , to the next one, x_{k+1} .



1.2 Picard's Method (Integration)

Picard's method helps to solve the (IVP) by a sequence of approximations as $y_i(x), i = 1, 2, \dots, k$ to the solution in which n^{th} approximation depends on the previous approximations. It is an iterative method used for approximating the solution of the (IVP). The solution will be more accurate if it is used repeatedly. It is easier to implement this method and the solution generally in power series.

According to the Volterra integral equation we have

$$y(x) = y_0 + \int_{x_0}^x f(\tau, y(\tau)) d\tau.$$

Now, by replacing $y = y_0$ as

$$y_1(x) = y_0 + \int_{x_0}^x f(\tau, y_0(\tau)) d\tau \text{ is the first approximation.}$$

The second approximation is

$$y_2(x) = y_0 + \int_{x_0}^x f(\tau, y_1(\tau)) d\tau.$$

By induction, it results the recursion of Picard's iterative formula

$$y_{k+1}(x) = y_0 + \int_{x_0}^x f(\tau, y_k(\tau)) d\tau, k = 0, 1, 2, \dots \quad (\text{Picard's iterations})$$

Remark 16 *Picard's method is of considerable theoretical value, but can be applied only to a limited class of equations in which the successive integrations can be performed easily. The method can be extended to simultaneous equations and equations of higher order.*

1.3 Taylor series method (Derivation)

Let $y(x)$ be the exact solution of equation (IVP). If the existence of all higher order partial derivatives is assumed for y at x_0 . Then the Taylor series for y around x_0 is given by

$$y(x) = y(x_0) + (x - x_0)y'(x_0) + \frac{(x - x_0)^2}{2!}y''(x_0) + \dots + \frac{(x - x_0)^n}{n!}y^{(n)}(x_0) + \dots$$

Since at x_0, y_0 is known, y' at x_0 can be found by computing $f(x_0, y_0)$. Similarly higher derivatives of y at x_0 also can be computed by making use of the relation $y' = f(x, y), y'' = \frac{\partial f}{\partial x} + f \frac{\partial f}{\partial y}, \dots$ and so on.



Hence the value of y at any neighboring point x_0 can be obtained by summing the above infinite series. However, in any practical computation, the summation has to be terminated after some finite number of terms. If the series has been terminated after the p^{th} derivative term then the approximated formula is called the Taylor series approximation to y of order p and the error is of order $p + 1$.

Example 17 Using Picard's method and Taylor series method to solve the following (IVP):

$$\begin{cases} y' = 1 + y = f(x, y) \\ y(0) = 0, x \in [0, 1] \end{cases}$$

a) Picard's Method : we have

$$y_k(x) = y_0 + \int_{x_0}^x f(\tau, y_{k-1}(\tau)) d\tau = \int_0^x (1 + y_{k-1}(\tau)) d\tau, k = \mathbb{N}^*;$$

So,

$$y_1(x) = \int_0^x f(1 + y_0) d\tau = \int_0^x d\tau = x;$$

$$y_2(x) = \int_0^x (1 + \tau) d\tau = x + \frac{x^2}{2};$$

$$y_3(x) = \int_0^x \left(1 + \tau + \frac{\tau^2}{2}\right) d\tau = x + \frac{x^2}{2} + \frac{x^3}{3!};$$

$$y_3(x) = \int_0^x \left(1 + \tau + \frac{\tau^2}{2} + \frac{\tau^3}{3!}\right) d\tau = x + \frac{x^2}{2} + \frac{x^3}{3!} + \frac{x^4}{4!};$$

$\vdots$

$$y_n(x) = \sum_{k \geq 1} \frac{x^k}{k!} \simeq e^x - 1.$$

b) Taylor series method :

$$y(x) = y(0) + xy'(0) + \frac{x^2}{2!}y''(0) + \dots + \frac{x^n}{n!}y^{(n)}(0) + \dots$$

we have

$$y(0) = y_0 = 0;$$

$$y' = 1 + y \implies y'(0) = 1 + v = 1;$$

$$y'' = (1 + y)' = y' \implies y''(0) = y'(0) = 1;$$

$\vdots$

$$y^{(n)}(0) = y'(0) = 1.$$



So,

$$y(x) = x + \frac{x^2}{2!} + \dots + \frac{x^n}{n!} + \dots = e^x - 1.$$

Remark 18 *This is a single step method and works well so long as the successive derivatives can be calculated easily. If $f(x, y)$ is somewhat complicated and the calculation of higher order derivatives becomes tedious, then Taylor series method cannot be used gainfully. This is the main drawback of this method and therefore, has little application for computer programs. However, it is useful for finding starting values for the application of powerful methods like Euler and Runge-Kutta, which will be described in the following section.*

